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Human Breast Cancer: Visium Fresh Frozen, Whole Transcriptome

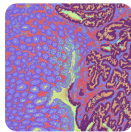
Spatial Gene Expression dataset analyzed using Space Ranger 1.3.0

Mapped
95.7%



Assess data quality

View summary metrics to assess data quality and more.

[View summary](#)

Visualize and explore data

Discover differentially expressed genes, visualize your favorite genes, and explore your data with our visualization software.

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10x Genomics obtained fresh frozen Invasive Ductal Carcinoma breast tissue from [BioIVT](#). The tissue was embedded and cryosectioned as described in Visium Spatial Protocols - Tissue Preparation Guide Demonstrated Protocol ([CG000240](#) Rev A). Tissue sections of 10 μm thickness were placed on Visium Gene Expression Slides.

The tissue was AJCC/UICC Stage T2N0M0, ER positive, PR negative, Hercep Test 2+.

- Block ID: 1168993F
- Tumor Grade III
- Sex: Female (Age 60s)
- RIN Score: 9.26

The slide was cover-slipped and the H&E image acquired using a Nikon Ti2-E microscope with the following settings:

- Color camera
- 20X objective
- Numerical Aperture: 0.75
- Exposure: 10 ms
- Gain: 4.5X

The library was prepared following the Visium Spatial Gene Expression Reagent Kits User Guide ([CG000239](#) Rev A).

- Sequencing instrument: Illumina NovaSeq 600, flow cell H2LJMDSXY (lane 1-3)
- Sequencing Depth: 47,223 reads per spot

Product

Spatial Gene Expression

Software

Space Ranger

Subpipeline

spaceranger count

Species

Human

Anatomical Entity

Breast

Preservation Method

Fresh Frozen

Disease State

breast cancer

Biomaterial Type

Specimen from Organism, Imaged
Specimen

Date Published

2022-07-01

- Sequencing Configuration: 28 x 120 bp
- Sequencing Coverage: Read 1 - 16 bp (includes 16 bp Spatial Barcode, 12 bp UMI); Read 2 - 120 bp (transcript); i7 sample index - 10 bp; i5 sample index - 10 bp.
- Dual-Index set: T1T2-E5
- Slide: V19B23-014
- Capture Area: A1

Key cell metrics were:

- Spots detected under tissue - 4,898
- Median UMI counts per spot - 9,720
- Median genes per spot - 3,654

To maintain donor anonymity, sequence data for this sample is not currently available.

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